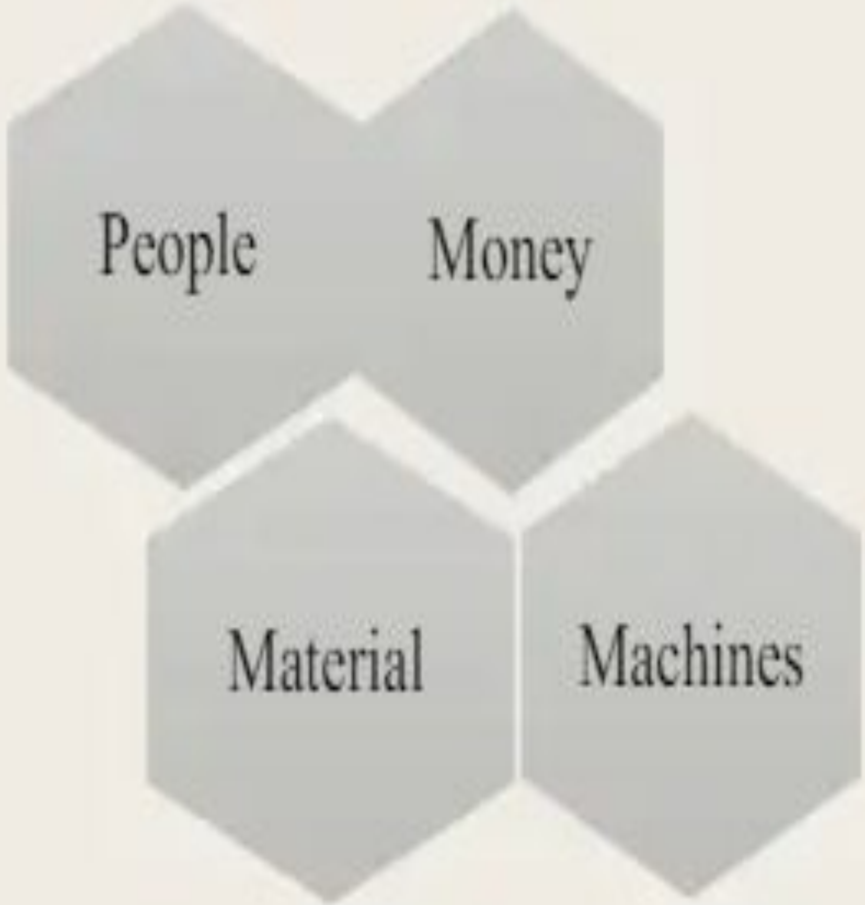
TECHNOLOGICAL INNOVATION MANAGEMENT AND ENTREPRENEURSHIP

21EC61

- Management is a critical element in the economic growth of a country
- Without management, a country's resources of production remain resources and never become production
- Management is essential in all organised efforts, be it a business activity or any other activity.
- Management is the dynamic, life giving element in every organisation.



George R Terry

Management as a process "consisting of planning, organising, actuating and controlling, performed to determine and accomplish the objectives by the use of people and resources"

Four Management Activities

Mary Parker Follett

“art of getting things done through people”

Two weaknesses of this definition are

- Art deals with the application of knowledge. Management also involves acquisition of knowledge i.e science. Management based on rules of thumb or intuition is not correct management

- This definition does not throw light on the various functions of a manager.

Planning

Managers think of their actions in advance, their actions are usually based on some method, plan or logic

Organisin
g

Managers coordinate the human and material resources of the organisation.

Actuating

Managers motivate and direct subordinates

Controllin
g

Managers attempt to ensure that there is no deviation from the norm or plan

Management Functions

Newman and Summer - organising, planning, leading and controlling.

Henri Fayol- planning, organising, commanding, coordinating and controlling.

Luther Gulick - "POSDCORB", planning, organising, staffing, directing, coordinating, reporting and budgeting.

Warren Hayens and Joseph Massie- decision making, organising, staffing, planning, controlling, communicating and directing.

Koontz and O' Donell - planning, organising, staffing, directing and controlling.

Management Functions

Planning

Organizing

Staffing

Directing(leading)

Controlling

Innovation

Planning

function performed by managers at all levels- top, middle and supervisory.

Plans made by top management for the organisation as a whole may cover periods as long as five or ten years.

Plans made by middle or first line managers, cover much shorter periods(next day's work or a two hour meeting) function performed by managers at all levels- top, middle and supervisory.

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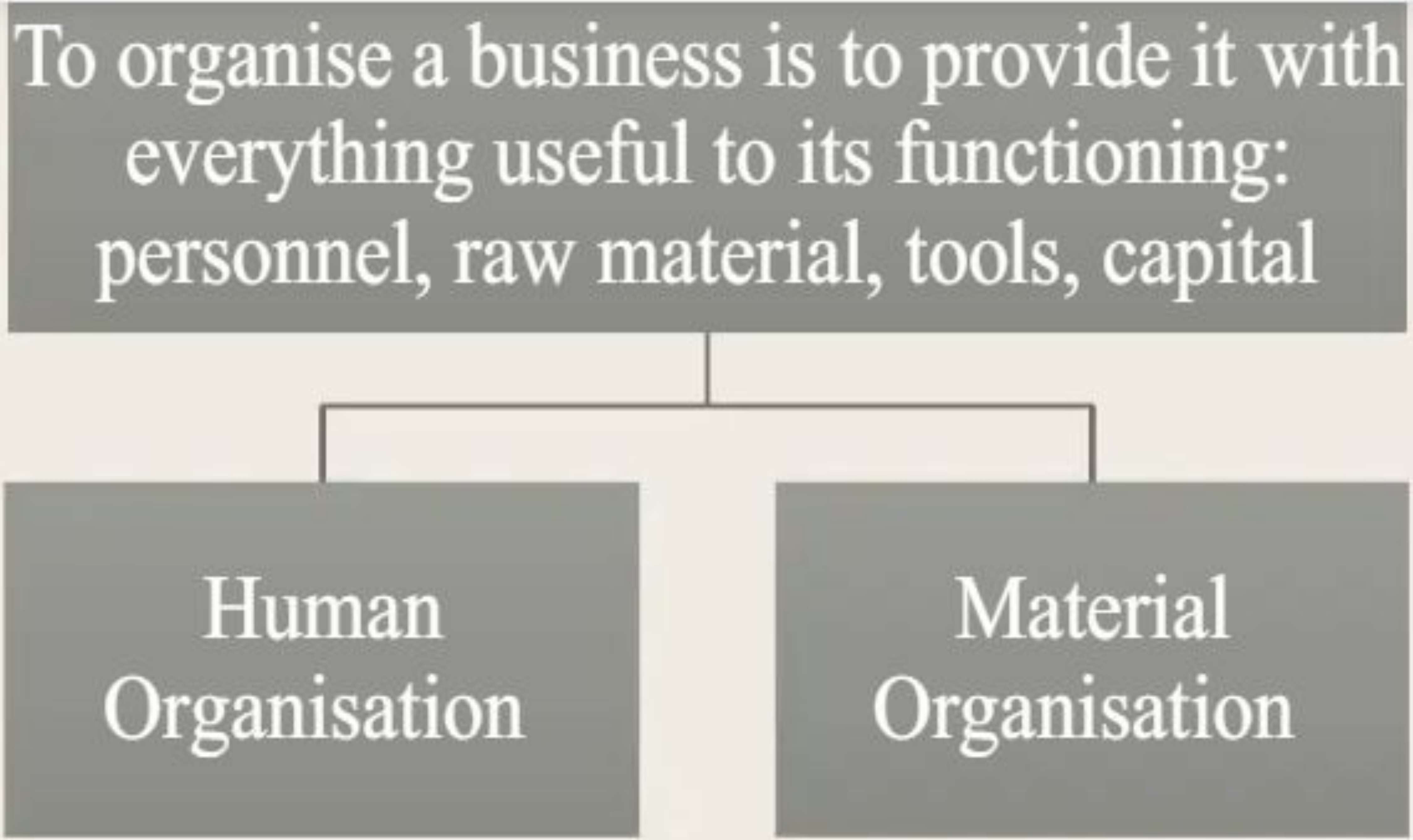

To organise a business is to provide it with everything useful to its functioning: personnel, raw material, tools, capital
Human Organisation
Material Organisation

Staffing

Find the right person for each job

Selection and training of the future managers and encouraging highly disciplined approach to work

Suitable system of compensation.

Directing


Directing
Motivation
Financial
Non financial
Leadership
Manager guides and influences
Communication
Passing information and understanding

Manager must ensure that

- Everything occurs in conformity to plans adopted.
- Whether instructions are used correctly
- Whether principles are established as planned.

Controlling Functions

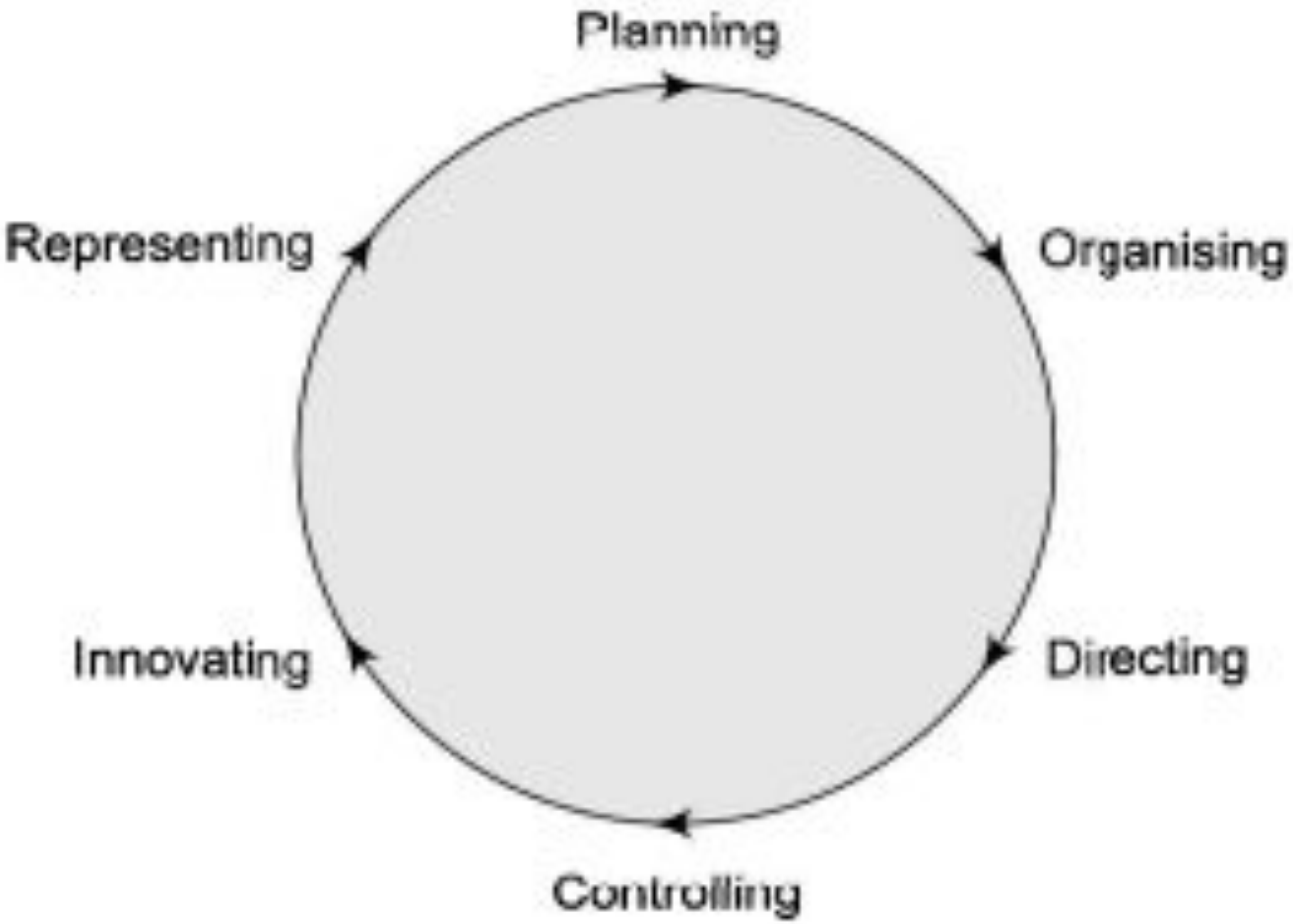
- Establishing the standards of performance
- Measuring current performance and comparing against the established standards.
- Taking action to correct any performance that does not meet the standards.

Innovation

Creating new ideas which may either result in the development of new product or finding new uses of old ones. It is not for an organisation to grow bigger, but it is necessary that it constantly grows better.

REPRESENTATION

- ✓ Manager has to represent his organisation to the stake holders, government officials, labour unions, financial institution, supplier, customer etc.
- ✓ Manager must win their support by effectively managing the social impact of his Organisation.


Planning
Organising
Directing
Controlling
Innovating
Representing

LEVELS OF MANAGEMENT

- The three levels of management that are commonly found in an organization are:

TOP MANAGEMENT:

- Top management consist of chairman, directors, company presidents, vice presidents CEO’s.
- Functions are:
- They determine goals of organization
- Establish policies
- Monitoring performances
- Judging the results

MIDDLE MANAGEMENT:

- It is a vast & diverse group that includes finance manager, sales manager, marketing manager, personal manager departmental heads etc.
- Functions are:
- To plan details of all operations. • To co-ordinate between various departments • Development of manpower for the organization by imparting training.

LOWER MANAGEMENT:

- It includes foreman & supervisors. They are basically one step above the workers.
- Functions are:
- To act as link between management & the workers.
- Supervision of workers & their work. • Arrangement of materials & tools etc. for production

Representative organisational or business functions

Management functions	Production ↓	Sales ↓	Finance ↓	Personnel ↓
Planning →				
Organising →				
Directing →				
Controlling →				
Innovating →				
Representing →				

ROLES OF MANAGER

- According to Mintzberg , manager should be regarded as playing the following 10 different roles:
- 1.Inter personal roles: FIGUREHEAD, LEADER, LIAISON. 2.Informational roles: MONITOR, DISCMINATOR, SPOKESMAN.

3.Decisional roles: ENTERPRENEUR, DISTRBANCE HANDLES, RESOURCE ALLOCATER & NEGOTIATOR

INTERPERSONAL ROLES

FIGUREHEAD:-In this role, every manager will perform some duties of ceremonial nature. Such as greeting and touring dignitaries, attending wedding of an employee, taking important customer to lunch & so on.

LEADER:-As a leader, every manager must motivate & encourage his employees. He should reconcile the needs with the goals of the organization.

LIASION:-In this role, every manager must cultivate contacts with outside people and collect useful information for the well being of the organization.

INFORMATIONAL ROLES:

MONITOR: As monitor, manager has to solicits & receives information from various sources on matters concerning his organization. Keeps all channel open for information. DISSEMINATOR:-The manager passes some of his privileged information directly to his key subordinates, who has no access otherwise.

SPOKESMAN:-As a manager, he should spend part of his time in representing his organization before various outside groups. Eg: share holders, govt. bodies, press media etc

DECISIONAL ROLES:

ENTREPRENEUR:-As an entrepreneur, a manager initiates projects to improve the organization, encourages new ideas and innovations, brings in changes & manages it looks for sources.

DISTURBANCE HANDLER:-A manager handles all kinds of disturbances that occur within the organization & find solutions to problems such as worker strikes, major accidents, customers going bankrupt, suppliers not delivering to schedule etc.

RESOURCE ALLOCATOR:-Here manager decides how all the organizations get distributed among various divisions or employees and allocates to required areas. NEGOTIATOR:-Manager need to negotiate on important matters concerning the organization. Eg: negotiating trade unions, politicians, client, suppliers etc.

MANAGERIAL SKILLS

Manager should possess 3 major skills:

TOP MANAGEMENT—CONCEPTUAL SKILL

MIDDLE MANAGEMENT—HUMAN RELATION SKILL

SUPERVISORY LEVEL—TECHNICAL SKILLS

The skill is an individual's ability to perform physical or mental tasks with a specified outcome.

CONCEPTUAL SKILL:

It is the ability of a manager to conceptualise the environment, the organization & his own job, so that he can set appropriate goals.

For his organization, for himself & his team this skill seems or increase in importance as manager moves up to higher positions of responsibility in the organization

TECHNICAL SKILL:

It is manager understands of the nature of job that people under him have to perform.

It refers to person's knowledge and proficiency in any type of process or technique.

This skill is important at lower levels of management. As manager moves to higher positions, the conceptual component becomes more important & technical component becomes less important

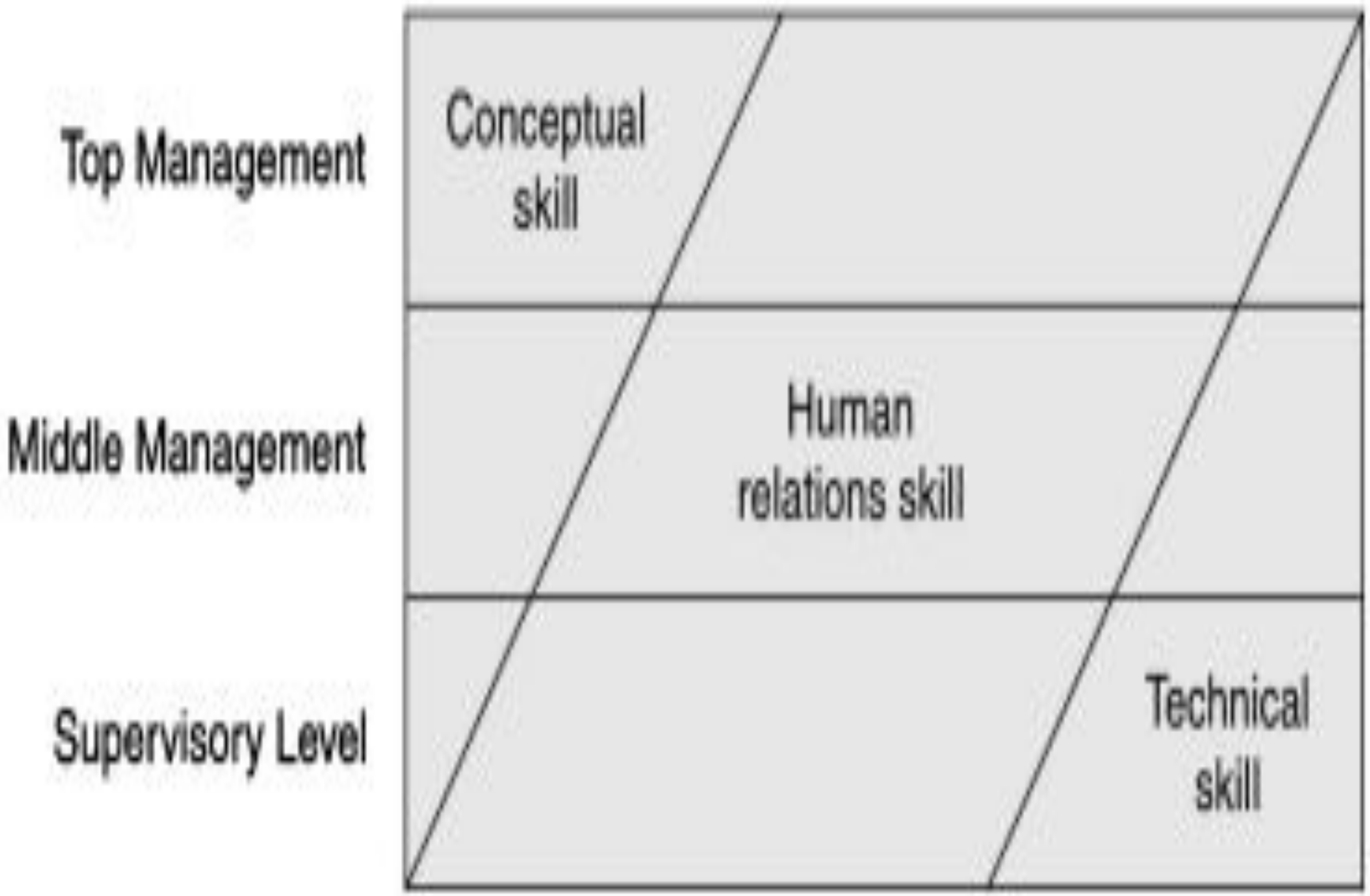
HUMAN RELATIONS SKILL:

It is the ability to interact effectively with people at all levels. This skill helps manager.

To judge the possible reactions to & outcomes of various course of action he may undertake to recognize the feelings & sentiments of others.

To examine his own concepts & values, this will enable manager to develop more useful attitudes about him.

This skill is important to managers at all levels. At top level technical skill becomes least important.


Top Management
Middle Management
Supervisory Level
Conceptual skill
Human relations skill
Technical skill

Planning

- Planning is deciding in advance what is to be done.
- It includes the selection of objectives, policies, procedures and programs from alternatives." -M. L. Harley
- "Planning is decision making in anticipation." -R.L. Ackoff

Planning is the *beginning* of the process of management. A manager must plan before he can possibly organise, staff, direct or control. Because planning sets all other functions into action, it can be seen as the most basic function of management. Without planning other functions become mere activity, producing nothing but chaos. This has been called the *principle of primacy of planning*.

Nature

NATURE OR FEATURES OR CHARACTERISTICS OF PLANNING

Planning is an intellectual process:

• It requires managers to think before acting.

• It is thinking in advance.

• It is by planning managers of organization decides what is to be done, when it is to be done, how it is to be done and who is to do it

Planning is decision making:

- Decision making is an integral part of planning.
- It is a process of choosing among alternatives.

  For ex:- planning for

  their organization, managers must decide which goal to pursue.

  shall we manufacture all parts internally or buy some from

  outside?”
- Managers must decide how they will allocate their resources to

  attain their goals.

Planning is a continuous process:

- Manager should constantly watch the progress of his plans
- He should constantly monitor the conditions, both within and outside the organization to determine if changes are required in his plans.
- Manager should be right than being consistent.
- It is called “ principle of navigational change.”

Planning must be flexible:

- Planning must not be rigid.
- Flexibility of planning means its ability to change direction to adapt to changing situations.
- Planning should possess a built –in flexibility in atleast 5 major areas.
- 1) Technology 2) Market 3) Finance 4) personnel 5) organization

Planning is all pervasive:

• Planning is important to all managers regardless of their level in the organization.

i.e from CEO to first time supervisor.

• Top level managers are concerned with longer time periods (six months to five years later).

• Lower level managers are concerned with planning activities for the day, week or month

IMPORTANCE AND PURPOSE OF PLANNING PROCESS.

Without planning business decisions would become random, ad hoc choices.

Four concrete reasons for the importance of planning function are as follows:-

1) Minimizes risk of uncertainty.
2) Leads to success.
3) Focuses attention on the organization’s goals.
4) Facilitates control and trains executives.

Planning minimizes risk and uncertainty:-

Organizations have to function and survive in an environment that is changing too fast.

In such situations, planning based on hard facts and and data help managers to minimize risk and uncertainties

Planning leads to success:-

Planning leads to success by doing beyond mere adaptation to market fluctuations.

With the help of a sound plan, management can act proactively, and not simply react

Planning focuses attention on the organizations and goals:-

- Planning helps the manager to focus attention on the organization’s goals and activities.
- This makes easier to co-ordinate the resources of the organization more economically.
- Planning helps manager to think in advance and orderly sequence of steps for realization of organization’s goals and to avoid needless overlapping of activities.

Planning facilitates control:-

- The function of control is to ensure that the activities confirm the plans.
- Without proper control planning is a wasteful exercise

Planning trains executives.

- Planning is also an excellent means for training executives.
- Then executives become involved in the activities of the organization

Planning ensures co-ordination:-

- Planning helps to establish coordinated efforts from different divisions, departments and people.
- When all the members of the organization become aware of goals and the routes taken to achieve them, there is better co ordination

Planning encourages innovation and creativity:-

- In order to overcome weakness and ward off threats, managers are compelled to be creative, innovative all the time.

- This will help managers to remain competitive at all times

Planning helps in decision making:-

- Managers in all levels in an organization are continuously included in decision making in one way or the other.
- Planning helps in providing guidelines and thus facilitate decision making

Planning improves morale:- (team spirit)

- Planning has a definite role to play in the success of an organization.
- Success leads to profit and profit leads to happy management.
- Because of the fact that employees have participated in the planning process, it will boost their morale

In a large organisation, there are various types of plans that are arranged in a hierarchy within the organisation. This means that plans at each level have to be consistent with and contributive to the achievement of plans above them (Fig. 4.1). We now describe each type of plan in detail.

Fig. 4.1 Hierarchy of organisational plans

TYPES OF PLAN

VISION:

- This is the dream that an entrepreneur creates about the direction that his business should pursue in future.
- Vision is an ongoing process.
- As the organisation proceeds, vision reshapes.
- A vision should be brief, focused, clear & inspirational to the organisational employees.
- It should be linked to customer's need & convey a general strategy for achieving the mission

Mission:

- It is the unique aim of an organisation that sets apart from others of its types.
- Mission decides the scope of business [i.e products & services the organisation provides]
- mission is not revised every now and then in response to every new turn in the economy
- it may change over time to take advantage of new condition

OBJECTIVES:

- objectives are the goals or aims that the management wishes the organisation to achieve in pursuit of its mission.
- It is the end points towards which all business activities like organising, staffing, directing & controlling are directed

CHRACTERISTICS OF OBJECTIVES:

- objectives are multiple in number
- objectives are either tangible or intangible
- objectives have a priority
- objectives are generally arranged in a hierarchy
- objectives may clash with each other

Objectives are multiple in number:

- all organisations usually have multiple objectives
- It is because of the many areas of business operations where organisations try to set their goals.
- "peter ducker" says, there are 8 areas in which objectives to be set

(a) Market standing of the company

(b) innovation achieved by the company

(c) productivity in various parameters

(d) physical & financial resources

(e) profitability of business

(f) managerial performance & development

(g) worker performance and attitude

(h) fulfilment of public responsibility

Objectives are either tangible or intangible

- It is tangible ,which means they could be quantifiable ex: productivity, profitability... etc
- objectives are intangible which means they are qualitative in nature and cannot be quantified for ex: workers morale, company image.... etc

Objectives have a priority:

- At any point of time , accomplishment of one objectives is relatively more important than of the others.
- The establishment of priorities is extremely important because in that resources of any organisation must be allocated by rational means.
- At all time ,managers are confronted with alternative goals which must be evaluated and ranked .ex: university vice chancellor must determine the relative importance of teaching, research & extension goals

Objectives are generally arranged in hierarchy

- It means the top level objectives of an organisation break into next level of objectives which again give rise to many more in the next level so on
- therefore objectives form a hierarchy from the top level to the lowest position in the organizational structure

Objectives sometime clash with each other

The process of allocating objectives among various unit creates the problem of potential goal conflict, where in achieving the goal of one unit may harm or fail other goals.

ex: goal of production unit is to achieve low cost unit or price by producing in mass.

Mass production should not be of low quality. Because, it might conflict with the sales goal of selling high quality product

STRATEGIES

- Is term originated in military, which connates a response to a competitive environment this is commonly referred to as SWOT (strength, weakness, opportunities & threats) analysis.
- Two important activities involved in strategy formulation are environmental appraisal & corporate appraisal.
- Environment Appraisal: An analysis of the relevant environment result in the identification of threats and opportunities. Some of the key factors are: Political and legal factors, Economic factors Competitive factors, Social and cultural factors
- Corporate Appraisal: This involves an analysis of the company's strengths and weaknesses.

Strategic planning	Operational planning
1. It decides major goals and policies of allocation of resources to achieve these goals.	1. It decides the detailed use of resources for achieving these goals
2. It is carried at higher level of management.	2. It is carried at lower level of management.
3. It is long term.	3. It is short term.
4. It is based on long term forecasting considering the possible impact of political, economical, technological and competitive factors and is more uncertain.	4. It is generally based on past performance of the organization and is less uncertain.
5. It is less detailed.	5. It is more detailed.

SINGLE USE PLANS

They are developed to achieve a specific end after reaching that target ,that plan becomes useless.

The major types of plans are programmes and budgets Programmes: they are precise plans or definite steps in proper sequence which need to be taken to discharge a given task

Budgets: budgets are plans for a future period of time containing statements of expected results in numerical terms i.e rupees ,man-hours, product -units & so on

STANDING PLANS

They are designed for situations that often repeat. These plans are used again.

There are 4 types of standing plans : (1) Policies (2) Procedures

(3) Rules (4) Methods

Policies: setting up boundaries that supply the general limits & direction in which managerial action will take place

Procedures: are detailed guidelines that are used to carry out the policies.

Rules: are, detailed & recorded instruction that a specific action must meet under the given instruction. ex: reporting time to office ,lunch time ...etc

Methods: method is a way in which one step of a procedure to be carried out

Policy	Procedure
General guidelines of the organization.	General guidelines at the action level.
Top level activity.	Departmental activity.
Policies fulfill the objectives of an organization	Procedures guide the way to implement the policies.
Policies are often made without any study or analysis.	Procedures are always made after through study and analysis of work.

STEPS OF PLANNING

1. Establishing verifiable goals or set of goals to be achieved
2. Establishing planning premises ➡ Internal and external premises ➡ Tangible and intangible premises ➡ Controllable and non controllable premises
3. Deciding the planning period
4. Finding alternative courses of action
5. Evaluating and selecting a course of action
6. Developing derivative action plans
7. Establishing and developing action plan
8. Measuring and controlling the process

1. Establishing verifiable goals or set of goals to be achieved

- The first step in planning is to determine the enterprise objectives
- It is set by upper level or top level manager
- The type of goal selected will depend on a number of factors : the basic mission of the organization, value its manager hold, and potential activities of the organization

2. Establishing planning premises

It is second step in planning.

It is to establish planning premises.

i. e certain assumptions about future on the basis of which the plans will be ultimately formulated.

Planning premises can be classified into a) Internal and external premises b) Tangible and intangible premises c) Controllable and non controllable premises

a) Internal and external premises :

Internal premises within the organization. some of the examples are policies, forecast, investment, availability of equipment, funds flow.. Etc

External premises are outside the organization they include government policies, technology changes, business environment, demand etc.

b) Tangible and intangible premises

Tangible are measurable premises. Ex : population, investment and demand. Intangible premises are which cannot be quantitatively measurable. Ex: business environment, economic conditions.

c) Controllable and non controllable premises : Some of the premises are controllable like, technical man power, input technology, machinery etc. Some other premises like strikes, non availability of raw materials, phase shifts are uncontrollable by the organization

3) Deciding planning period

- Once upper level managers have selected basic long-term goals and planning premises, the next task is to determine the period of the plan.
- In some stances plans are made for a Year only while in other plans spans decides choice of period are as follows

a. Lead time in development and commercialization of new product . b. Pay back period etc.

4) Finding alternative courses of action:

The fourth step in planning is to search and examine alternative courses of action. Ex: technical know how may be taken by engaging or foreign technician or by training staff abroad.

5) Evaluating and selecting a course of action :

Fifth step is to evaluate the alternative courses in the light and select the best of action

This is done with the help of quantitative technique and Operating research

6) Developing derivative plans :

- Once the plan for the organization has been formulated, middle and lower level managers must draw up the appropriate plans for their sub units.
- Derivative plans are plans that are required to support the basic plan
- In developing these derivative plans, lower level manager take steps similar to those taken by upper level manager. Selecting realistic goals etc

7) Establishing and deploying action plans

- Managers may not know how to turn the derivative plans into action.
- Action plan identifies particular activities necessary for this purpose and specifies the who, what, when and how of each action item

8) Measuring and controlling the progress :

- Without monitoring plan will get ruined
- Hence process of controlling is critical part of any plan. Managers need to check the progress of their plans.

LIMITATIONS OF PLANNING

- Planning is an expensive and time consuming process. It includes money, energy and risk, but gives no assurance of the fulfilment of organisation's objectives.
- Planning restricts the organisation to risk-free opportunities. Planning forces manager to operate within the limits, planning cause delay in decision making.
- For industries producing fashionable articles or for industries engaged in a publication of textbooks, working on a day-to-day basis is more economical than on a planned basis

- Establishment of advance plans may make administration inflexible. When unforeseen changes in the environment such as business recursion, crop failure take place then original plan loses its value and fresh plan need to be done.
- There is difficulty in formulating accurate premises, because future cannot be known with accuracy.
- Planning May sometimes face people’s resistance to do it.

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- When unforeseen changes in the environment such as business recursion, crop failure take place then original plan loses its value and fresh plan need to be done.
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DECISION MAKING – MEANING

A decision is a choice between two or more alternatives. This implies three things,

1. When manager make decision they are choosing.
2. Managers have alternatives when they are making decision. [Evaluate several alternatives and select the best one].
3. Managers have a purpose in mind when they are making a decision.

TYPES OF DECISION MAKING

- Programmed and non-programmed decision.
- Major or minor decision.
- Routine and strategic decision.
- Individual and group decision.
- Simple and complex decision

Programmed and non-programmed decision

- Programmed decisions are those that are made in accordance with some policy, rule or procedure and they do not have to be handled each time they occur. They are repetitive routine. Example: determining salary payment to employs who have been ill.
- In programmed decision, each manager is guided by some set of rules and policies. Therefore it is not possible for two different managers to arrive at different solution.

- Non programmed decisions are natural and non repetitive. If a problem has not arisen before then it must be handled by a non-programmed decision. Example: What to do about a failing product line, how community relations should be improved etc..
- In case of non programmed decisions, each manager will be his own personal beliefs, attitudes and value judgment to bear on decision process. Therefore two different managers can arrive at two different solutions.

Major or minor decision

- Degree of futurity of decision
- How long a decision in future will commit the company?
- A decision which has long range impact like replacement of men by machinery must be rated as major decision.
- How to store raw materials may be taken as minor decision
- Recurrence of decision
- Decisions which are rare considered as major decisions and decisions which occur very often is minor decision

- Impact of the decision on other functional areas
- If a decision affects only one function then it is a minor decision else it is a major decision. Qualitative factors that enter the decision
- A decision which involves certain subjective factors is an important decision
- Example: Subjective factors are basic principles of conduct, ethical values etc

Routine and strategic decisions

- Routine and tactical or housekeeping decisions are those which are supportive. They relate to the present.
- Their purpose is to achieve as high degree of efficiency possible in companies ongoing activities Example: providing Air conditioning (ac), better lighting etc.
- Lowering price of product, changing the product line are strategic decisions

Individual and group decisions

- Decision may be taken as either by an individual or by a group.
- Individual decisions are taken where problem is of routine nature
- Important and strategic decisions may result in some change in organisation is a group decision

Some *advantages* of group decisions are:

1. ***Increased acceptance by those affected*** Decisions made by a group are mostly accepted by the group's members, and they help implement those decisions more readily.

2. ***Easier coordination*** Decisions made by groups reduce the amount of coordination necessary to bring the decision into play.

3. ***Easier communication*** Decisions made by groups reduce the amount of communication necessary to implement the decision.

4. ***More information processed*** Because many individuals are involved, more data and information can be brought to bear on the decision. This can help improve the quality of the decision and uncover obstacles in the way of its execution.

Following are the *disadvantages* of group decisions:

1. ***Group decisions take longer*** Groups take longer than individuals to make decisions.

2. ***Groups can be indecisive*** Groups can drag on and never take decisions because they can always blame other members of the group for lack of progress.

3. ***Groups can compromise*** This can lead to decisions that satisfy the "lowest common denominator". It can lead to "*group think*" or conformity to peer pressure and neglect of better solutions.

4. ***Groups can be dominated*** The highest status individual, if he chooses, can influence the group so that it notices his or her choices. This negates the advantages of group decision-making.

5. ***Groups may have a prior commitment to a particular solution*** This may be due to ties to persons outside the groups, "empire building" attempts, or belief that a decision will have sufficient personal impact.

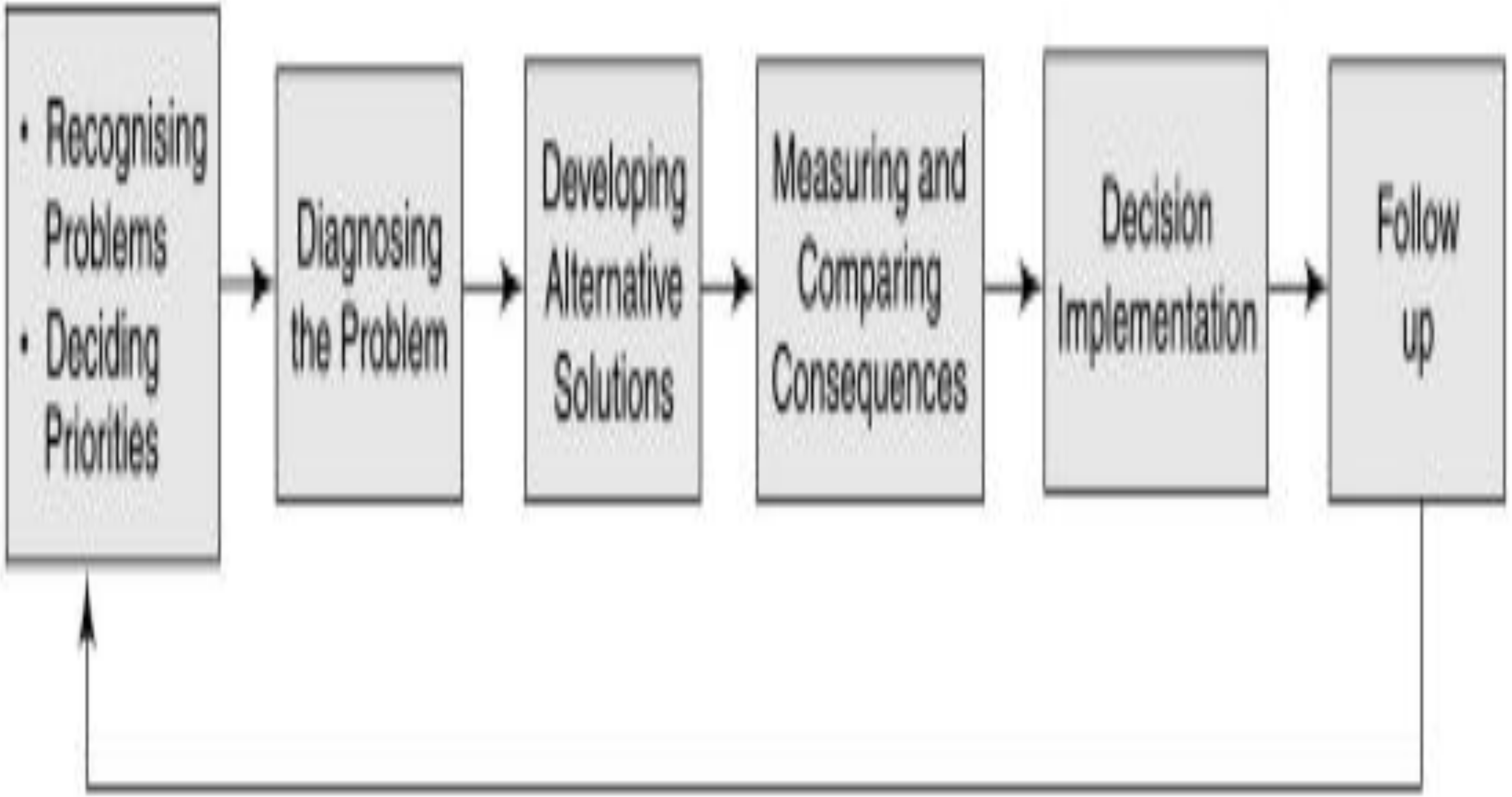
Simple and Complex decisions

When variables to be considered for solving a problem are few, the decision is simple; when they are many, the decision is complex. When we combine these two types of decisions with the low or high certainty of their outcomes, we get four types of decisions:

1. Decisions in which the problem is simple and the outcome has a high degree of certainty. These are called *mechanistic* or *routine* decisions. Managers often develop standard operating procedures to take these decisions.
2. Decisions in which the problem is simple but the outcome has a low degree of certainty. These are called *judgemental decisions.* Many decisions in the area of marketing, investment and personnel are of this type. For example, a marketing manager may have several alternative ways of promoting a product, but he may not be sure of their outcomes. Good judgement is needed for such decisions.
3. Decisions in which the problem is complex but the outcome has a high degree of certainty. These are called *analytical decisions.* Many decisions in the area of production are of this type. A variety of computational techniques, such as linear programming, network analysis, queuing theory, etc., are used to arrive at such decisions.
4. Decisions in which the problem is complex and the outcome has a low degree of certainty. These are called *adaptive decisions.* Changes in corporate plans and policies to meet the changes in environment and technology are decisions of this type. These decisions usually require the contributions of many people with diverse technical backgrounds, and may even need frequent modifications to adapt to the changing environment.

STEPS IN DECISION MAKING

- There are seven steps in decision making
- Recognizing the problem.
- Deciding Priorities among problems
- Diagnosing the problem.
- Developing alternative solutions or course of action.
- Measuring and comparing sequences of alternative solutions
- Converting the decision into effective action
- Follow-up



• Recognising Problems
• Deciding Priorities
Diagnosing the Problem
Developing Alternative Solutions
Measuring and Comparing Consequences
Decision Implementation
Follow up

Recognizing the Problem

- It is the first step in decision making
- A problem exists:
- When there is deviation from past experience.
- When there is deviation from plan
- When other people bring problems to the manager

1. *When there is a deviation from past experience* For example, this year's sales are falling behind last year's; expenses have suddenly increased; employees' turnover has grown; too many defective products are suddenly coming off the assembly line.
2. *When there is a deviation from the plan* For example, profit levels are lower than anticipated; a department is exceeding its budget; a project is off schedule.
3. *When other people bring problems to the manager* For example, customers may complain about late deliveries, a lower level manager may complain about the high performance standard set for him, workers may complain about poor working conditions and so on.
4. *When competitors outperform the manager's organisation* For example, other companies might develop new processes or improvements in operating procedures.

Deciding Priorities among problem:

- Manager should identify the problem which
- He can solve
- His subordinates can solve.
- Are to be referred to the higher officers

A manager should not allow himself to be bogged down by all sorts of problems. On examination he will find that some of his problems are such which can be solved best not by him but by his subordinates because they are closest to them. All such problems should be passed on to the subordinates. Then, some problems may need to be referred upward because they affect other departments or require information available only at a higher level. Some problems may be deferred because it may not be the best time to act. For example, in many product development areas, delaying a decision may be advisable until the outcome of anticipated technical breakthrough is known. Some problems may even be procrastinated and allowed to be solved without any effort. The classic example of this principle concerns Napoleon, who was reputed to let incoming mail pile-up on his desk for three weeks or more. When he finally read the accumulated mail, he was pleased to find that most matters had been resolved in the interim. Ultimately, there would be only very few problems requiring the manager's attention.

Diagnosis problem

- Manager should follow systems approach in diagnosing a problem.
- He should make a thorough study of all sides of a problem along with organization before arriving at a solution.

Every problem should be correctly diagnosed. A manager should remember that the symptoms of a problem that he observes may sometimes mislead him. They may lead him to suspect one part of the system or operation when, in fact, the defect may lie hidden in another part which is perhaps less visible. For example, if the sales decline, the management may think that the problem is one of poor selling procedures, sharp competition, or the saturation of old markets. But the real problem may be the tightfisted control of the firm and its consequential inability to move quickly enough to meet changes in customers' demand. Hence the efforts made by the management in other directions may not help in increasing sales.

Development of Alternate Solution

- After diagnosis next step is to find alternate solutions.
- Alternates do exist. In the absence of past history of alternate solutions, managers should find alternatives of his own

if a factory manager is considering a proposal to raise production to meet increased demand, there are several alternatives which can be considered. These might be: (i) build a new plant, or (ii) buy better equipment, or (iii) add an extra shift, or (iv) authorise overtime.

While developing various alternatives, a manager should always keep in mind their feasibility. He should have clear recognition of all those *limiting factors* which can make the accomplishment of an alternative difficult or impossible. Thus, in the above example, various limiting factors might be non-availability of cash and credit, late delivery of capital equipment, anti-expansion laws, or hostile attitude of employees.

Measuring and comparing Sequences

This comparison involves comparing a quality and acceptability of these alternatives

Decision implementation: This requires the communications of the decisions to the concerned employees in clear and simple terms.

Follow up: In final step, the action should be continuously followed up to ensure whether the decision is achieving its purpose.

The manager has to carry out the follow up action. If the result is not satisfactory, the manager has to take necessary corrective action or modify his decision.

Some of the difficulties faced by manager are: Incomplete Opposition by subordinates, improper communication, Wrong timing, Statutory regulations, Government policies, External influences and Lack of support.

Once appropriate alternative solutions have been developed, the next step in decision-making is to measure and compare their consequences. This involves a comparison of the *quality* and *acceptability* of various solutions. The quality of a solution must be determined after taking into account its tangible and intangible consequences. Tangible consequences are those which can be quantitatively measured or mathematically demonstrated. For example, one can easily calculate the costs of the various alternatives to meet rising demand in the foregoing example and compare the results. If analysis shows that the demand is only temporary, it would not be worthwhile to build a new plant or even to purchase new equipment. Thus adding a second shift or authorising overtime might be the most profitable course. Using overtime would have the advantage of making an immediate increase in production possible, although it would, of course, require extra labour costs and this would have to be taken into account. Intangible consequences are those which cannot be quantitatively measured or mathematically demonstrated. For example, one cannot easily calculate and compare the effects of good labour relations in one industrial location against low local taxes in another or of selecting a supplier with a lower price proposal against another with a better reputation for meeting delivery promises. *One must then use his judgement and intuition to solve the problem.* There is nothing wrong in relying on one's judgement in such a situation. On the contrary, it is harmful to manufacture data to facilitate some calculations.

The next step is to translate the decision into action. A decision is not complete until someone has been assigned responsibility to carry it out. This requires the communication of decision to the employees in clear and unambiguous terms. Failure to inform everyone who needs to know of the decision can lead to unfortunate results. Consider the problem of a food company that decided to launch an extensive promotional campaign on one of its speciality items. The management carefully prepared the sales force

and had newspaper, magazine and television advertisements ready. However, they did not properly inform the production group. As a result, sufficient amounts of the product were not available to take full advantage of the increased demand generated by the promotional campaign. By the time production rates could be increased, much of the potential benefit of the campaign was lost.